4. Inductance and Capacitance

Introduction to Electric Circuits

Supplemental Instruction

Cooper McAllister

April 2 2026

1 The Inductor

The inductor is a circuit element indicated by either of the following symbols:

I will use the left symbol here.

An inductor has an **inductance** L, measured in Henries (H). Expressed in SI units, the Henry is

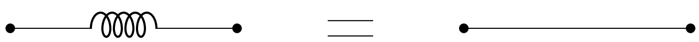
$$H = \frac{V \cdot s}{A} = \frac{\Omega \cdot s}{rad} = \Omega \cdot s$$

The equation relating voltage and current in an inductor is

$$v = L\frac{di}{dt}$$

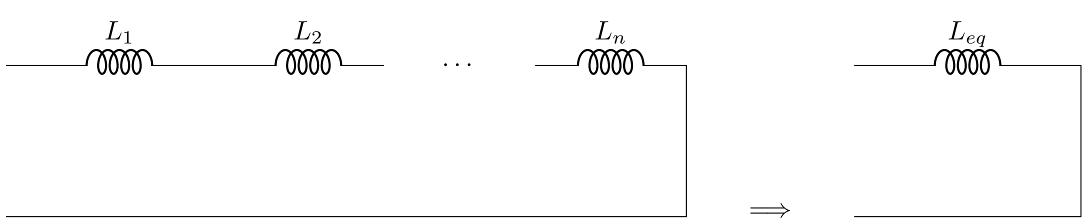
Current cannot change instantly in an inductor. This is because for current to change instantly, its derivative would approach ∞, which would mean voltage would have to approach ∞, which is impossible.

In steady-state (i.e. when nothing in the circuit is changing over time), an inductor behaves as a short. This is because the derivative of current is 0, so the voltage across the inductor is 0.

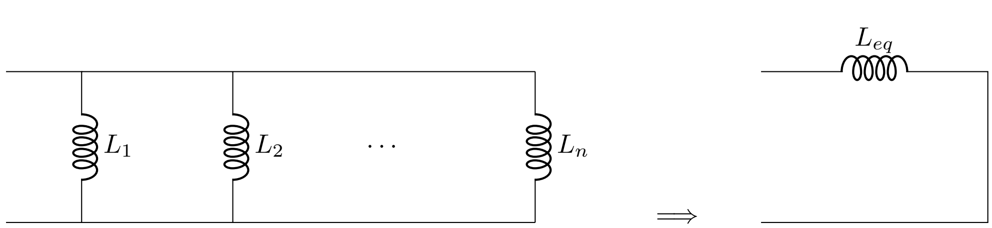


(steady state)

Inductors can be combined in parallel and series using the same formulas as for resistors:



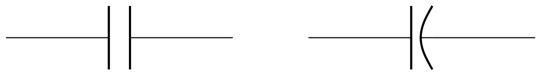
$$L_{eq} = L_1 + L_2 + \cdots + L_n$$



$$L_{eq} = \frac{1}{\frac{1}{L_1} + \frac{1}{L_2} + \cdots + \frac{1}{L_n}}$$

2 The Capacitor

The capacitor is a circuit element indicated by either of the following symbols. The right symbol is used for polarized capacitors, which are not discussed in this course.



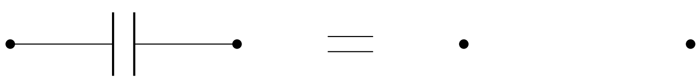
A capacitor has a **capacitance** C, measured in Farads (F). Expressed in SI units, the Farad is

$$F = \frac{A \cdot s}{V} = \frac{s}{rad \cdot \Omega} = \frac{s}{\Omega}$$

The equation relating voltage and current in a capacitor is

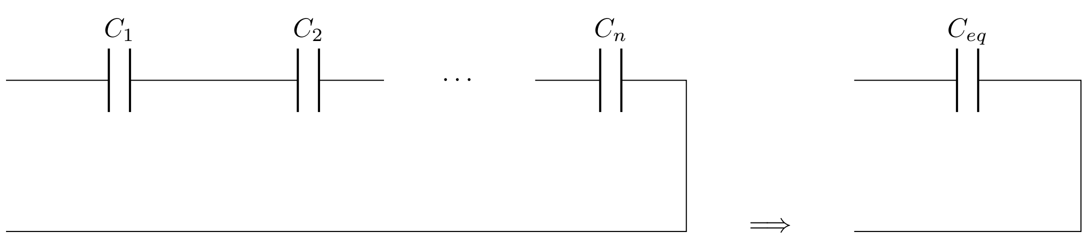
$$i = C\frac{dv}{dt}$$

Voltage cannot change instantly in a capacitor. This is because for voltage to change instantly, its derivative would approach ∞, which would mean current would have to approach ∞, which is impossible.
In steady-state, a capacitor behaves as an open circuit. This is because the derivative of voltage is 0, so the current through the inductor is 0.

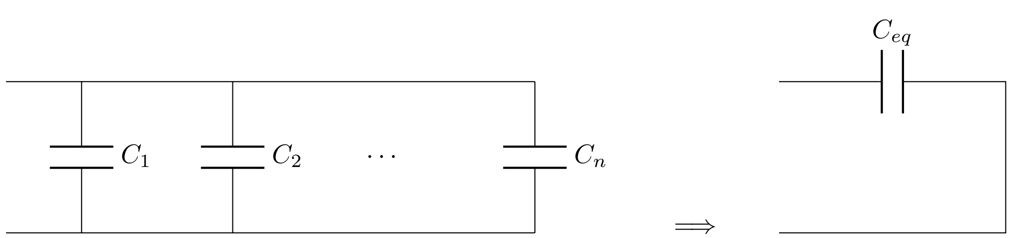


(steady state)

Capacitors can be combined in parallel and series using the same formulas as for resistors, but using the opposite formulas for series and parallel:



$$C_{eq} = \frac{1}{\frac{1}{C_1} + \frac{1}{C_2} + \cdots + \frac{1}{C_n}}$$



$$C_{eq} = C_1 + C_2 + \cdots + C_n$$

3 First-Order Time Response

Circuits with DC sources and *only one* capacitor or inductor exhibit first-order response. If a change takes place at time $t = 0$ (such as a switch opening or closing), any current or voltage in the circuit, $m(t)$, can be found as:

$$m(t) = (m(0^+) - m(\infty))e^{\frac{-t}{\tau}} + m(\infty)$$

where
$m(0^+)$ is the quantity evaluated immediately after the change takes place,
$m(\infty)$ is the quantity evaluated a very long time after the change takes place, and
τ is the time constant:

For a circuit with a capacitor,

$$\tau = R_{th}C$$

For a circuit with an inductor,

$$\tau = \frac{L}{R_{th}}$$

R_{th} is the thevenin resistance of the circuit as seen from the capacitor or inductor.
Note that $t = 0^-$ is used to denote the time immediately before the change takes place (while the circuit is at DC), $t = 0^+$ is used to denote the time immediately after the change takes place (the start of the response), and $t = \infty$ is used to denote the time a very long time after the change takes place (when the circuit is once again at DC).
A general procedure for finding the response of capacitor voltage or inductor current in a first-order circuit is as follows:

1. Draw the circuit before the change takes place (at $t = 0^-$) and replace capacitors and inductors with their DC equivalent.

2. Find:

   - For a circuit with a capacitor: $V_C(0^-)$, the voltage across the capacitor. Then, $V_C(0^-) = V_C(0^+)$ (because capacitor voltage cannot change instantly).
   - For a circuit with an inductor: $I_L(0^-)$, the current through the inductor. Then, $I_L(0^-) = I_L(0^+)$ (because inductor current cannot change instantly).
   - Note that $V_C(0^-)$ or $I_L(0^-)$ may be given if there is an initial capacitor voltage or inductor current.

3. Draw the circuit after the change takes place, replace capacitors and inductors with their DC equivalent, and find $V_C(\infty)$ or $I_L(\infty)$.

4. Find the time constant using the cirucit after the change takes place.

5. Substitute $m(0^+)$, $m(\infty)$, and τ with the proper values in the above equation to obtain the time response.

References

- M. Iordache. Class Lectures, Topic: “Electric circuits.” EEGR 2053, School of Engineering and Engineering Technology, LeTourneau University, 2024.
- M. Iordache. 2024. EEGR 2053 Electric circuits [PowerPoint slides]. Available: https://mviordache.name/EEGR2053/index.html
- T. L. Floyd and D. M. Buchla, Principles of Electric Circuits: Conventional Current, 10th ed. Hoboken, NJ: Pearson Education, 2020.
- O. Ortiz. 2025. ENGR 2053 Intro to electric circuits [PowerPoint slides].